

Assignment 1

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1.a True

$$\lim_{N \rightarrow \infty} \frac{2N+5}{N} = 2$$

1.b False

$$\lim_{N \rightarrow \infty} \frac{0.01N}{N^{0.99}} = \infty$$

1.c False

$$\lim_{N \rightarrow \infty} \frac{2^N}{2^{\frac{N}{2}}} = \lim_{N \rightarrow \infty} 2^{\frac{N}{2}} = \infty$$

1.d True

$$\lim_{N \rightarrow \infty} \frac{\ln(N)}{\sqrt{N}} = \lim_{N \rightarrow \infty} \frac{\frac{1}{N}}{\frac{1}{2\sqrt{N}}} = \lim_{N \rightarrow \infty} \frac{2\sqrt{N}}{N} = 0$$

1.e True

$$\lim_{N \rightarrow \infty} \frac{N \ln^2(N^2)}{N^2 \ln(N)} = \lim_{N \rightarrow \infty} \frac{2 \ln(N)}{N} = 0$$

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2.a

$$O(N^2)$$

2.b

$$O\left(\frac{N^2}{2}\right) = O(N^2)$$

2.c

$$O(N)$$

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3.a

$$d1 = \text{high} - \text{low}$$

$$p = \text{low} + \frac{\text{high} - \text{low}}{2} = \text{low} + \frac{d1}{2} = \frac{\text{high} + \text{low}}{2}$$

3.a.a

$$d2 = \text{high} - (p + 1)$$

$$\lim_{d1 \rightarrow \infty} \frac{d1}{d2} = \lim_{d1 \rightarrow \infty} \frac{d1}{\text{high} - \text{low} - \frac{d1}{2} - 1} = \lim_{d1 \rightarrow \infty} \frac{d1}{\frac{d1}{2} - 1} = 2$$

3.a.b

$$d2 = p - 1 - \text{low}$$

$$\lim_{d1 \rightarrow \infty} \frac{d1}{d2} = \lim_{d1 \rightarrow \infty} \frac{d1}{\text{low} + \frac{d1}{2} - 1 - \text{low}} = \lim_{d1 \rightarrow \infty} \frac{d1}{\frac{d1}{2} - 1} = 2$$

3.a.c

Each time sqrt is recursively called it splits the amount of numbers to search in half.

3.b

$$a^r = \text{high} - \text{low} \rightarrow 2^r = \text{high} - \text{low} \rightarrow \log_2(\text{high} - \text{low}) = r$$

$$r = \frac{\log(\text{high} - \text{low})}{\log(2)}$$

3.c

$$N = \text{high} - \text{low}$$

$$\text{tightest big O: } \log_2(N) = \frac{\log(N)}{\log(2)} \rightarrow \log(N)$$

3.d

$$\text{tightest big O: } \log_2(N) = \frac{\log(N)}{\log(2)} \rightarrow \log(N)$$

This is because at each call of the function it stores a constant amount of data (x, low, high, p) and if the function is called $\log_2(N)$ times then that will also be the space complexity.

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4.a

$$O(\log_3(N)) = O\left(\frac{\log(N)}{\log(3)}\right) \rightarrow O(\log(N))$$

4.b

$$O(2^{x+y})$$